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Real-Time Sign Language Translation to American English with different YOLO models

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Abstract. Sign language serves as a vital mode of communication for millions of Deaf and hard-of-hearing people, relying on hand gestures, facial expressions, and body movements to convey meaning. However, its limited understanding among the general population often creates communication barriers. This study focuses on the real-time recognition of static American Sign Language (ASL) gestures using advanced YOLO models: YOLOv8, YOLOv9, YOLOv10, and YOLOv11. The models were tested on the ASL Alphabet dataset, which includes diverse hand gestures and variations to mimic real-world conditions. Among the models tested, YOLOv9 delivered the highest precision with an mAP score @50% of 0.967, demonstrating superior classification performance. This study emphasizes how innovations in YOLO model architectures enhance ASL gesture recognition by analyzing key metrics such as precision, recall, and mAP. These results highlight the transformative potential of AI-powered sign language translation systems to improve accessibility and foster inclusive communication in daily life and professional environments for the Deaf community.

Keywords: Sign Language Detection, YOLO Model, ASL Translation, ASL Alphabet Dataset

1 Introduction

Communication is an essential human requirement that lays the foundation for societal interaction. This enables individuals to express their thoughts, emotions, intentions and ideas, fostering collaboration, learning, and social bonding. For deaf and hard-of-hearing communities, sign language serves as a vital mode of expression, composed of hand gestures, facial expressions, and body movements. To address these challenges, there has been a growing interest in assistive technologies, particularly over the past decade, leading to the development of real-time sign language detection systems. These innovative systems aim to convert sign language into spoken or written language on the spot, enabling deaf and hearing

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individuals to communicate more seamlessly. Real-time sign language detection represents a compelling application of artificial intelligence (AI) and machine learning (ML) technologies, with the potential to revolutionize communication worldwide.

Sign language is not merely a collection of static hand gestures; each sign integrates a blend of movements, facial expressions, and body postures that are both contextually dependent and culturally specific. Furthermore, sign languages vary significantly across different regions and communities, with unique systems such as the American Sign Language (ASL)[1], British Sign Language (BSL)[2], and Indian Sign Language (ISL)[3], each featuring distinct vocabulary and grammatical frameworks. This diversity complicates the development of a universal sign language detection solution because each language presents unique challenges. For instance, ISL often requires the simultaneous use of both hands, adding further complexity to the detection efforts. In addition, individual signing styles, regional dialects, and personal characteristics contribute to the challenge of standardizing detection systems for diverse populations. Recent advancements in AI, particularly in computer vision and deep learning, have significantly bolstered the development of more sophisticated sign-language detection systems. Convolutional Neural Networks (CNNs) [4] have proven highly effective in recognizing patterns within visual data, playing a crucial role in enabling computers to decode hand gestures and facial expressions involved in signing. These networks are well suited for processing images and videos and are essential for analyzing the continuous movements that characterize sign language. Additionally, Recurrent Neural Networks (RNNs)[5] have been instrumental in handling sequential data, which reflects the series of gestures inherent in a sign language conversation. RNNs excel at capturing the temporal relationships between different video frames, thereby enhancing the comprehension of communication dynamics.

1.1 Literature Review

Sign language recognition has gained considerable attention in recent years, with advancements in machine learning and deep learning models improving accuracy and real-time performance. Various approaches have been proposed to bridge communication gaps for the deaf and hard of hearing.

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Table 1. Overview of Sign Language Recognition Studies.

Authors	Year	Dataset	Model	Accuracy
Sreemathy et al.[6]	2023	Proprietary static signs (80)	YOLOv4, SVM with MediaPipe	98.8% (YOLOv4), 98.62% (SVM)
Adeyanju et al.[7]	2021	Static ASL gesture (Kaggle)	KNN, MLP	95.81% (KNN), 96.96% (MLP)
Natarajan et al.[8]	2023	RWTH-PHOENIX-Weather 2014T, ISL-CSLTR, How2Sign	H-DNA (CNN, Bi-LSTM, GAN)	95%
Patel et al. [9]	2023	Proprietary dataset	SSD MobileNet V2, Faster R-CNN	80% (SSD MobileNet), 78% (R-CNN)
Adeyanju et al. [10]	2021	RVL-SLLL, ASLLVD	MLP-NN,RNN, LSTM, DeepConvLSTM	MLP-NN:94.06%, RNN:95.11%, LSTM:98%, DeepConvLSTM: 91.1%
Wadhawan et al. [11]	2020	ASL Alphabet, Sign Language MNIST, MS-ASL	CNN, Hybrid Models, Ensemble Learning	80%-95%
Sharma et al. [12]	2021	RWTH-PHOENIX-Weather 2014, CSL, MS-ASL	3D CNN Models, CNN (VGG, ResNet)	80%-92% (3D CNN), 75%-90% (CNN)
Bhatt et al. [13]	2022	AULTS	LSTM, GRU	97%

1.2 Motivation and Contribution

More recently, the emergence of transformer models has opened new avenues in sign language detection, building on their revolutionary impact in natural language processing. These sophisticated models excel at identifying patterns within complex, sequential data, making them particularly suitable for understanding the progression of signs over time. By incorporating transformer models, researchers have enhanced both the accuracy and speed of sign language detection systems, bringing us closer to achieving real-time translations. This research paper explores the current landscape of real-time sign language detection, focusing on the key technologies underpinning its development and the challenges that remain. We will evaluate the roles of CNNs, RNNs, and transformer models in improving the precision and efficiency of these systems while also discussing the impact of hardware advancements such as depth cameras and motion sensors. Additionally, we will investigate the potential applications of real-time sign language detection across different sectors, including education, healthcare, and public services. By offering a comprehensive overview of this field, our research

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seeks to clarify how close we are to achieving seamless, real-time communication for Deaf individuals and to outline the next steps necessary to make this vision a reality.

2 Methods and Materials

Over the years, several models like R-CNN, Faster R-CNN, and other similar methods have been used for detecting objects, including hand gestures. While they're pretty accurate, they tend to be slower and are not ideal for real-time applications like sign language recognition. In our work, we chose to use the YOLO (You Only Look Once) family of models—specifically YOLOv8, v9, v10, and v11—because they're much faster and work well for live detection. The YOLO model is a state-of-the-art real-time object detection system that has gained significant popularity in computer vision applications. Its single-stage detection architecture allows it to achieve superior speed and accuracy compared to traditional multi-stage detection models.

2.1 YOLOv8

Figure 1 showcases the architecture of YOLOv8 [14], pushing object detection to new heights with impressive gains in both speed and accuracy. Unlike its predecessors, it ditched the anchor box system, making it far more adaptable to different objects and environments. On top of that, it introduced advanced feature processing with the SPPF module, sharpening its ability to pinpoint objects with greater precision. The integration of CBAM in the neck further refined the model's ability to focus on relevant features, boosting performance. But what set YOLOv8 apart was how much more it could do beyond object detection.

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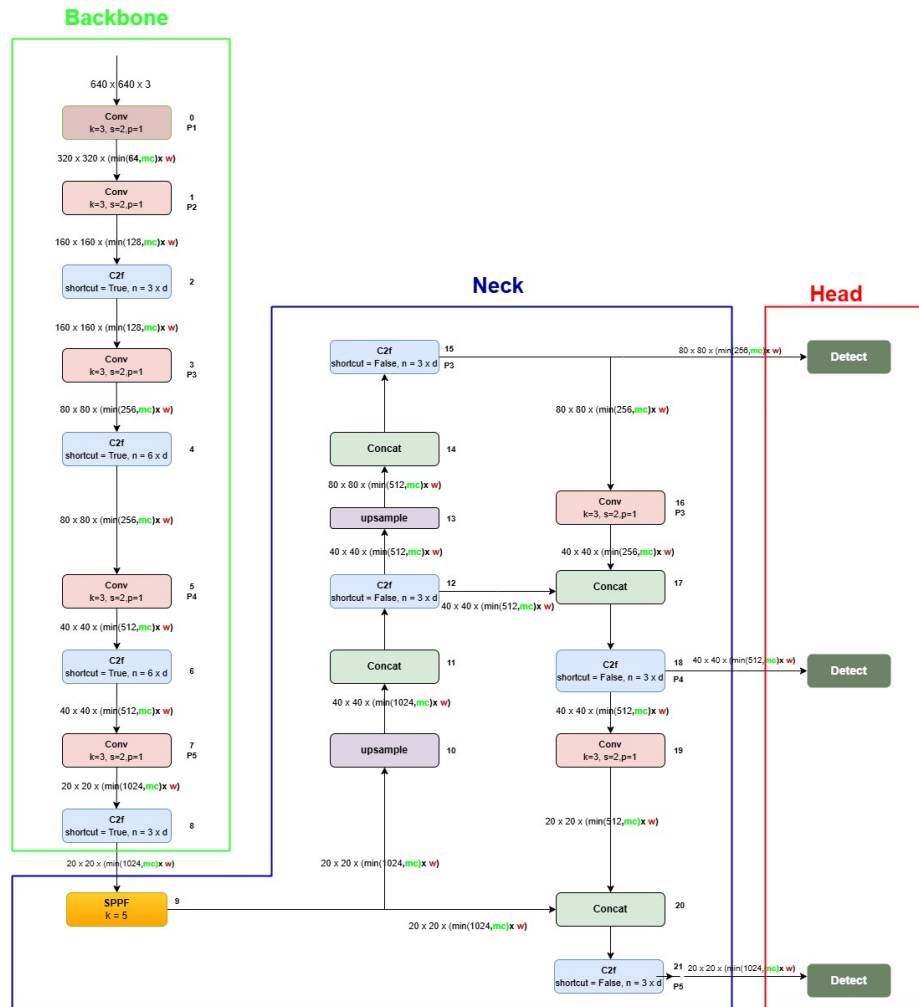


Fig. 1. YOLOv8 Architecture.

2.2 YOLOv9

Figure 2 showcases the architecture of YOLOv9 [15], took object detection to the next level by tackling a critical challenge—preserving vital details as data moved through the network. To achieve this, it introduced a groundbreaking method called PGI, which ensured that essential information wasn't lost in the process. Alongside this, it incorporated GELAN, a smarter approach to linking layers, making the system more efficient and precise. The integration of CBAM in the neck further refined the model's ability to focus on relevant features, boosting performance.

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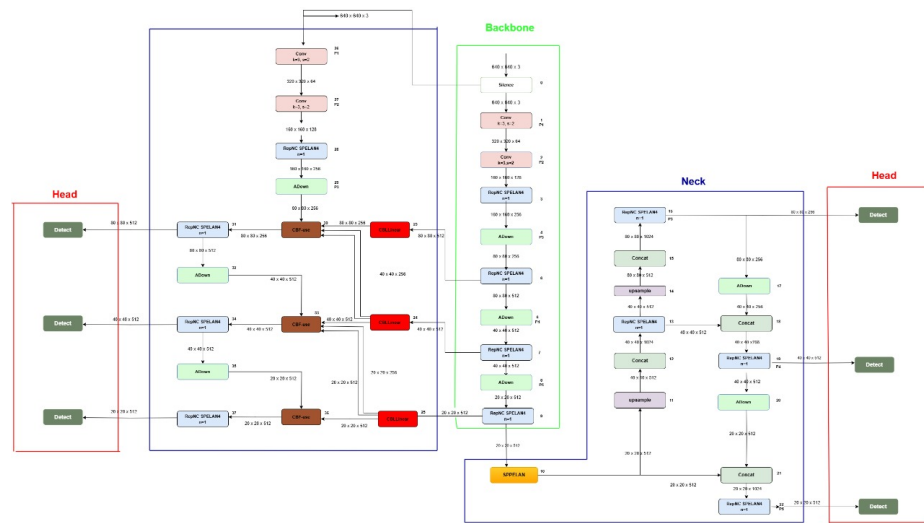


Fig. 2. YOLOv9 Architecture.

2.3 YOLOv10

Figure 3 showcases the architecture of YOLOv10 [16], took speed and efficiency to a whole new level. Instead of relying on the slower non-maximum suppression (NMS) method to filter out duplicate detections, it introduced a clever two-head system—one head for quick approximations and another for fine-tuning the results. This streamlined approach meant faster processing without compromising accuracy. The integration of CBAM in the neck further refined the model's ability to focus on relevant features, boosting performance. Another major upgrade was its optimized CSPNet-based feature extraction, which boosted performance while keeping things lightweight.

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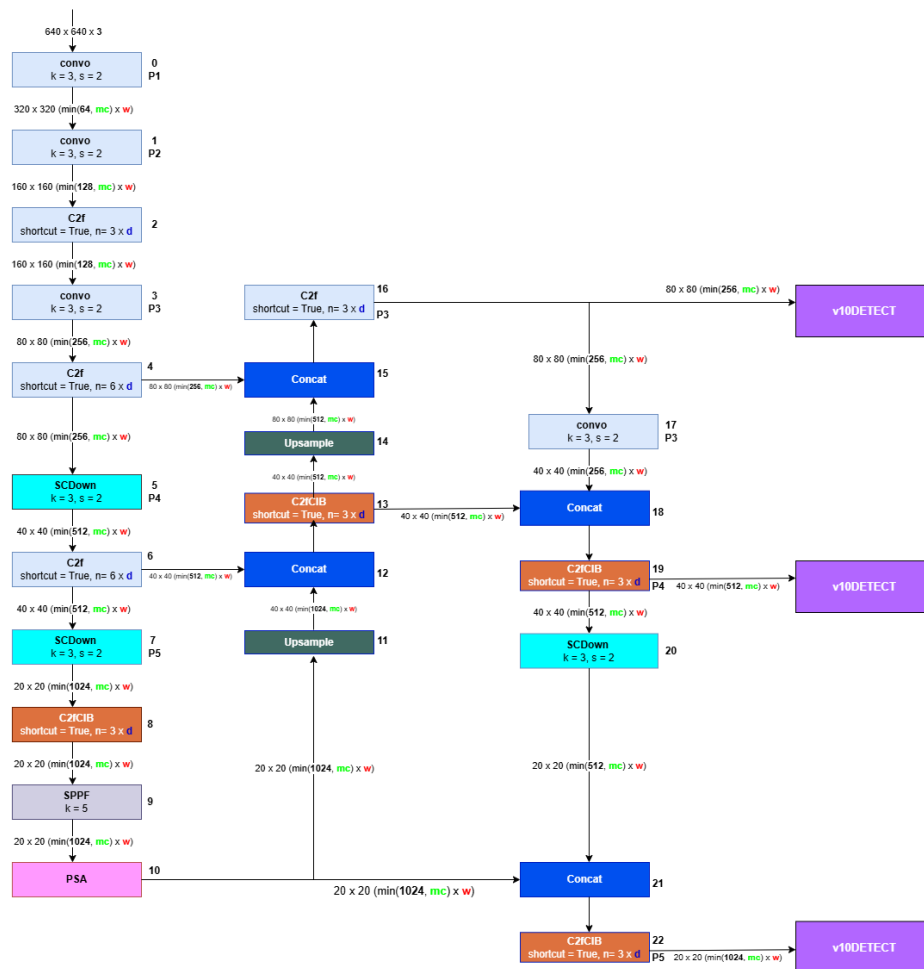
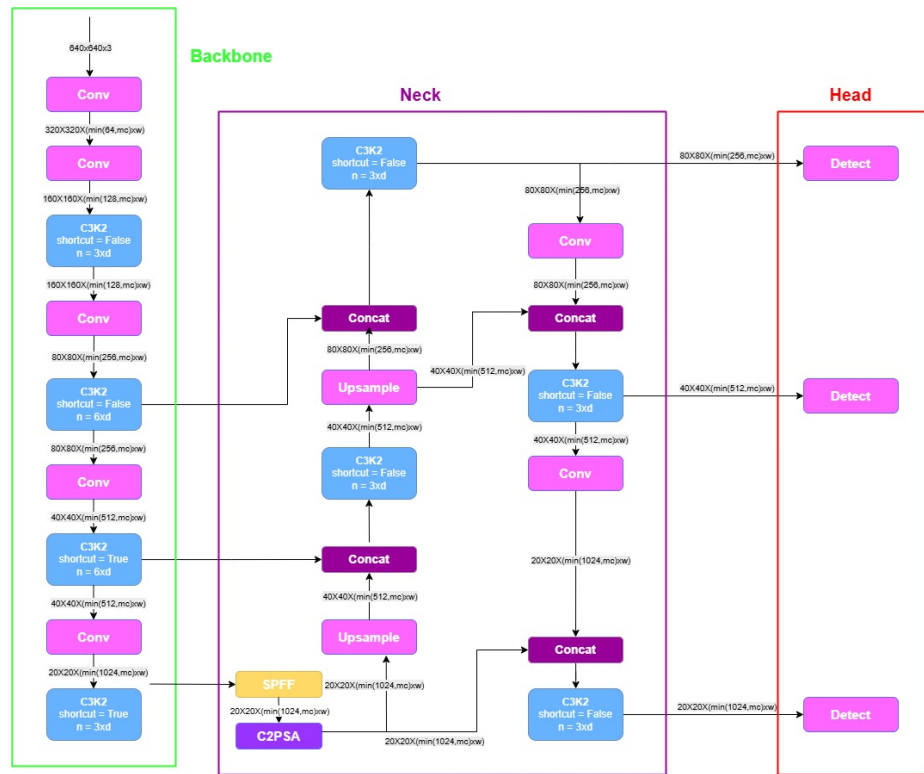


Fig. 3. YOLOv10 Architecture.

2.4 YOLOv11

Figure 4 showcases the architecture of YOLOv11 [17], embraced simplicity to achieve remarkable efficiency with less complexity. By streamlining its architecture and using 22% fewer components than YOLOv8, it managed to enhance its object detection capabilities significantly. This was made possible by its newly designed backbone and neck, which boosted both speed and accuracy. The integration of CBAM in the neck further refined the model's ability to focus on relevant features, boosting performance. What set YOLOv11 apart was its ability to take on more advanced challenges, such as pose estimation and detecting rotated objects (OBB).

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by gathering information across channels using both average-pooling and max-pooling. These operations create two separate 2D feature maps—one capturing overall patterns and the other emphasizing standout details.

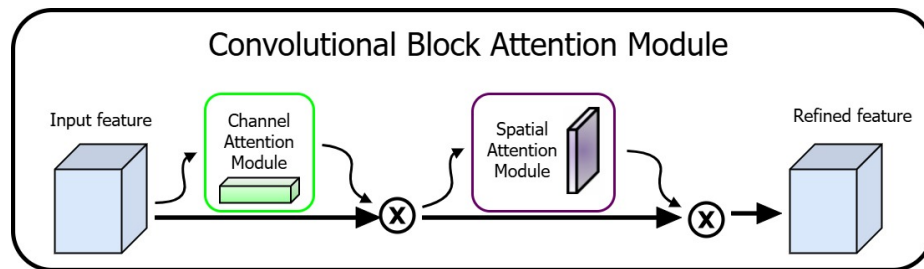


Fig. 5. The overview of CBAM.

3 ASL Alphabet Dataset

The ASL Alphabet Dataset contains about 87,000 RGB images which serve as an effective tool for building American Sign Language recognition models. The data set displays letters from A through Z and numbers between 0 and 9 together with SPACE, DELETE and NOTHING categories to help train models in identifying static hand symbols. The data set presents numerous images which demonstrate various hand poses across different background scenes and with different hand sizes thus permitting models to recognize signs under diverse circumstances.

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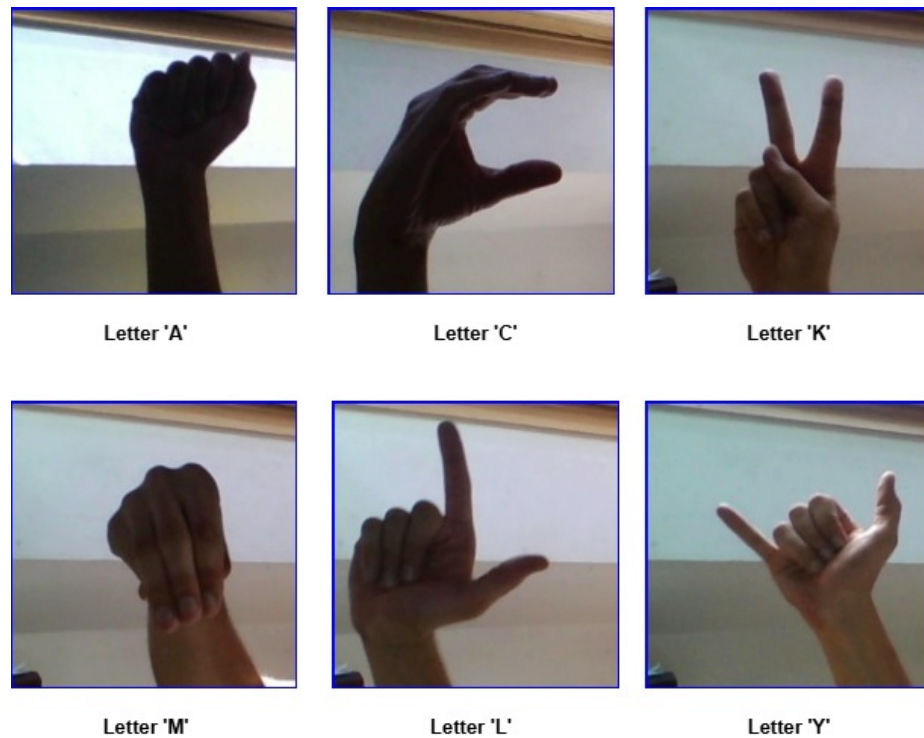


Fig. 6. Sample Image Representing from the ASL Dataset.

4 Results and Discussion

The evaluation metrics employed for assessing model performance included precision, recall, $mAP@0.5$, and $mAP@0.5:0.95$. Table 2 presents the results obtained from four YOLO model on the ASL Alphabet dataset. YOLOv9 achieved the highest accuracy with a $mAP@50\%$ of 0.967 and $mAP@50-95\%$ of 0.825. YOLOv11, while showcasing the second-highest classification accuracy, delivered slightly lower classification accuracy with a $mAP@50\%$ of 0.959, though still strong in real-time application scenarios. YOLOv10 and YOLOv8 offered a balanced trade-off between precision and latency, with consistent curve patterns across all three plots, indicating dependable performance. All four models were trained under identical conditions (50 epochs, same hyperparameters) to ensure fair comparison.

Figure 7 showcases sample test set outputs for ASL alphabet detection using the different YOLO versions. These visualizations highlight the models' effectiveness in distinguishing hand signs under varied lighting and background conditions.

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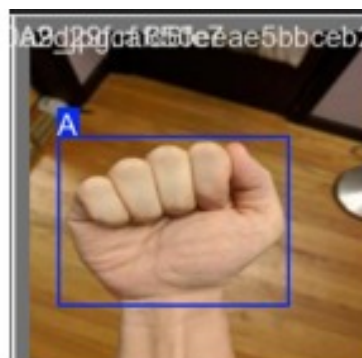
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Table 2. Results of YOLO models on the ASL Alphabet dataset.

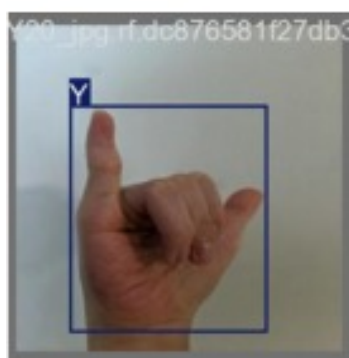
Model	Precision (P)	Recall (R)	mAP score @50%	mAP score @50-95%
YOLOv8 with CBAM	0.948	0.943	0.967	0.821
YOLOv9 with CBAM	0.948	0.925	0.967	0.825
YOLOv10 with CBAM	0.919	0.852	0.949	0.803
YOLOv11 with CBAM	0.886	0.887	0.959	0.816



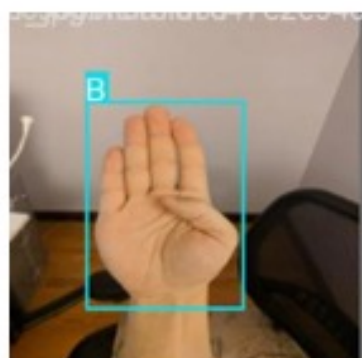
YOLO11



YOLO8



YOLO9



YOLO10

Fig. 7. Sample detection outputs using YOLOv8, YOLOv9, YOLOv10, and YOLOv11 on the ASL Alphabet dataset.

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4.1 Comparative Analysis

In this study, we analyzed the performance of four YOLO models—YOLOv8, YOLOv9, YOLOv10, and YOLOv11—on the ASL Alphabet dataset. To evaluate their classification capabilities, we examined precision-confidence, recall-confidence, and precision-recall (PR) curves, along with confusion matrices.

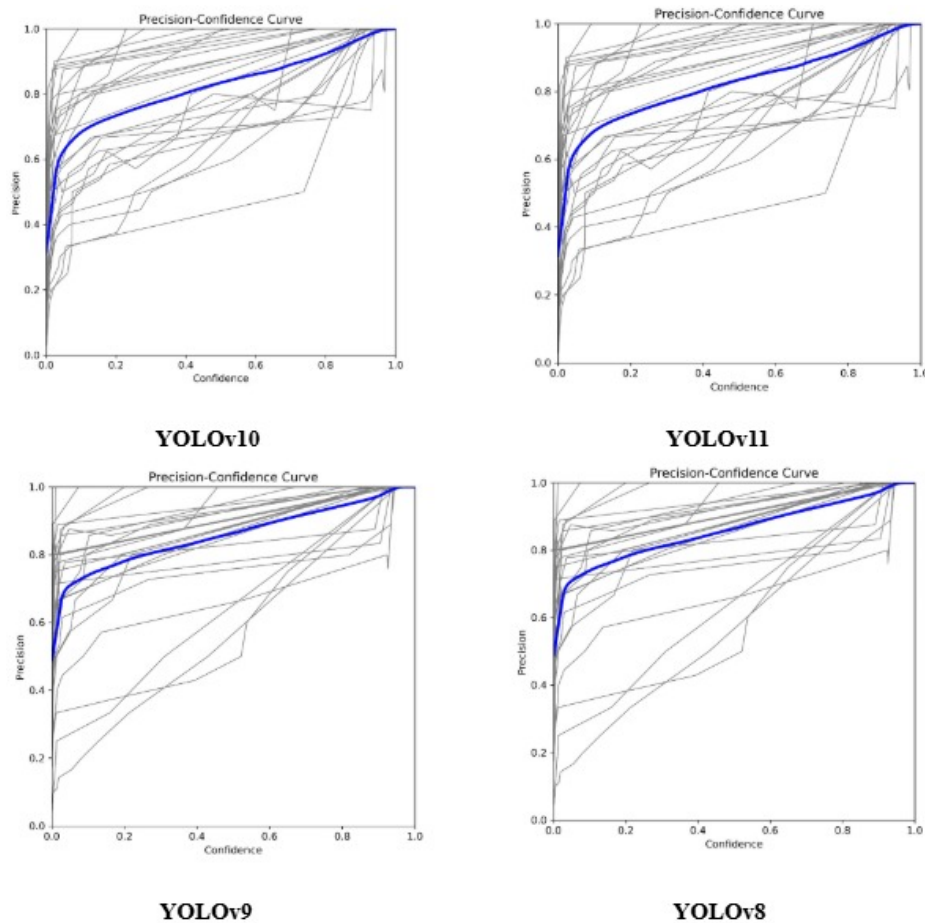


Fig. 8. Precision vs Confidence curve for different YOLO models

As shown in Fig. 8, YOLOv10 and YOLOv11 maintain high precision across increasing confidence thresholds, indicating strong prediction consistency. In contrast, YOLOv8 and YOLOv9 exhibit more variation, implying less reliability under high-confidence detections.

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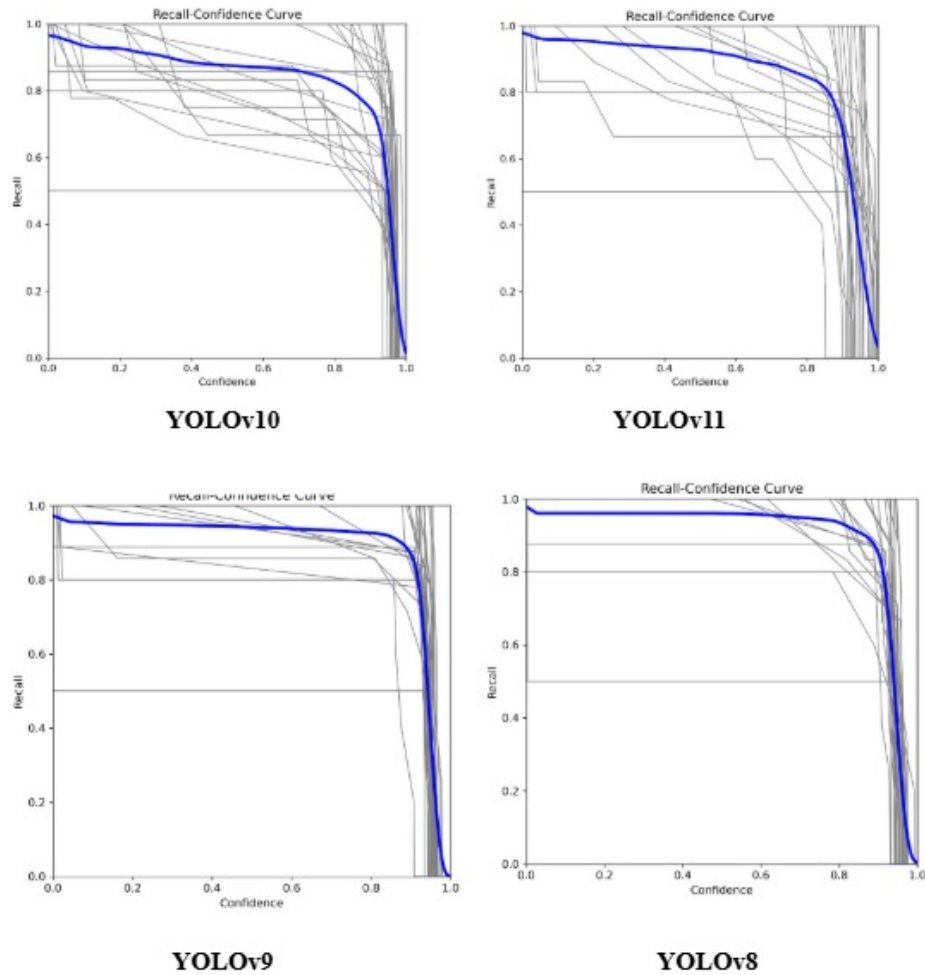


Fig. 9. Recall vs Confidence curve for different YOLO models

Fig. 9 reveals that both YOLOv10 and YOLOv11 sustain high recall values near 1.0 across a broad confidence range, reflecting their ability to detect true positives with minimal misses. YOLOv8 and YOLOv9 show earlier drop-offs, suggesting a higher miss rate at strict thresholds.

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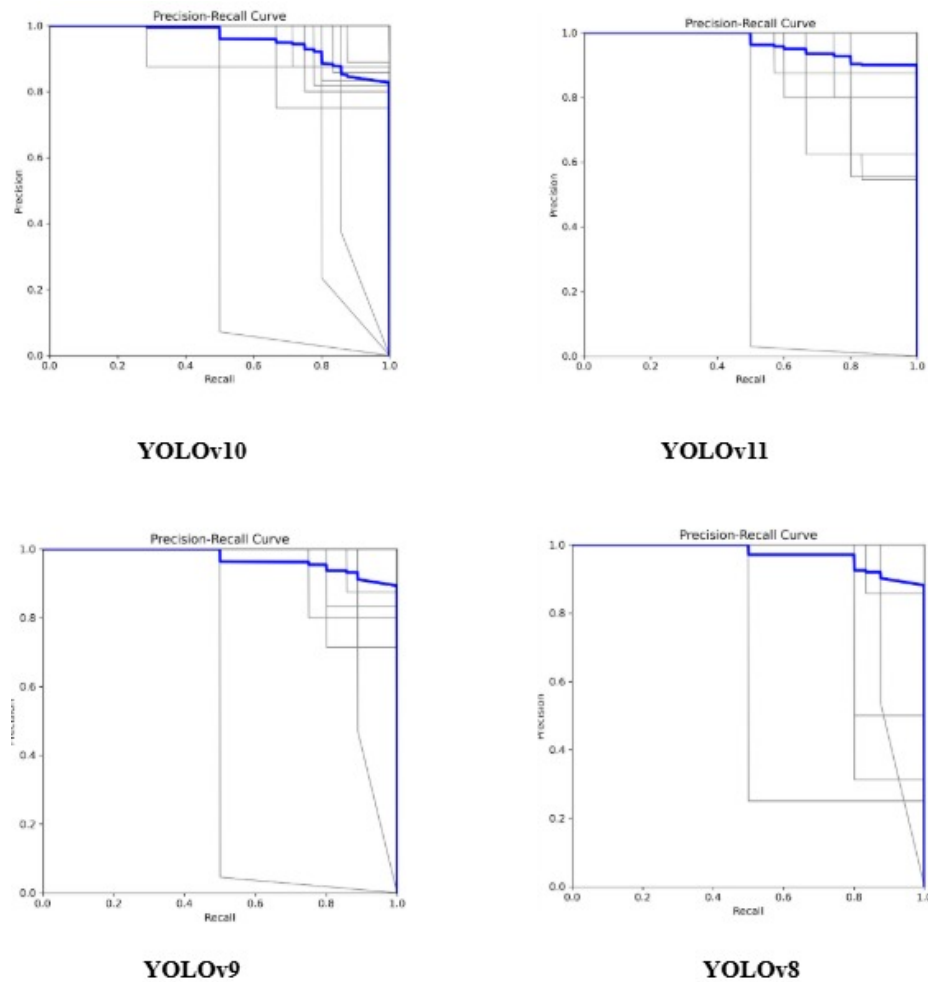


Fig. 10. Precision-Recall curves comparison

The PR curves in Fig. 10 indicate that YOLOv9 achieves the highest area under the curve (AUC), striking the best balance between precision and recall. YOLOv10 and YOLOv11 follow closely, while YOLOv8 trails with more fluctuating performance.

4.2 Discussion

In this study, four YOLO models are being tested - YOLOv8, YOLOv9, YOLOv10, and YOLOv11 on the ASL Alphabet dataset. The precision-confidence, recall-confidence, precision-recall (PR) curve, and confusion matrix analysis point to a significant variation in the accuracy and consistency of classification. YOLOv10

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and YOLOv11 demonstrate resilient precision across all confidence levels, making consistent predictions, while YOLOv8 and YOLOv9 are more variable. Similarly, YOLOv10 and YOLOv11 also have high recall values, minimizing missed detections compared to YOLOv8 and YOLOv9. PR curve analysis confirms YOLOv9's resilient precision-recall trade-off, while confusion matrices confirm YOLOv10 and YOLOv11's improved class-wise accuracy.

In general, the study identifies how architectural innovations in later iterations of YOLO enhance ASL recognition, which is valuable in practical applications.

5 Conclusion and Future Work

Four YOLO model variants—YOLOv8, YOLOv9, YOLOv10, and YOLOv11—for American Sign Language (ASL) gesture recognition are compared in this paper. To learn how various architectural modifications impact recognition accuracy and inference speed, we looked at important performance metrics like accuracy, recall, mAP@0.5, and mAP@0.5:0.95. Among all the evaluated models, YOLOv9 had the best classification accuracy, and YOLOv11 had the best inference efficiency. Our next goal is to recognise entire ASL sentences, going beyond static gestures. To capture movement over time, temporal models such as Transformers, GRU[19], and LSTM[18] must be incorporated. Incorporating hand tracking, facial expressions, and body posture analysis will also be necessary to improve recognition accuracy. We also intend to investigate real-time solutions that facilitate more seamless interaction methods for individuals who are Deaf and hard of hearing.

References

1. Valli, C., & Lucas, C. (2000). *Linguistics of American sign language: An introduction*. Gallaudet University Press.
2. Sutton-Spence, R., & Woll, B. (1999). *The linguistics of British Sign Language: an introduction*. Cambridge University Press.
3. Shenoy, K., Dastane, T., Rao, V., & Vyavaharkar, D. (2018, July). Real-time Indian sign language (ISL) recognition. In 2018 9th international conference on computing, communication and networking technologies (IC3-CNT) (pp. 1-9). IEEE.
4. Chauhan, R., Ghanshala, K. K., & Joshi, R. C. (2018, December). Convolutional neural network (CNN) for image detection and recognition. *Proceedings of the 2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC)*, 278–282. IEEE.
5. Cui, R., Liu, H., & Zhang, C. (2017). Recurrent convolutional neural networks for continuous sign language recognition by staged optimization. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7361-7369).

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6. Sreemathy, R., Turuk, M. P., Chaudhary, S., Lavate, K., Ushire, A., & Khurana, S. (2023). Continuous word level sign language recognition using an expert system based on machine learning. *International Journal of Cognitive Computing in Engineering*, 4, 170-178.
7. Adeyanju, I. A., Bello, O. O., & Adegboye, M. A. (2021). Machine learning methods for sign language recognition: A critical review and analysis. *Intelligent Systems with Applications*, 12, 200056.
8. Natarajan, B., Rajalakshmi, E., Elakkiya, R., Kotecha, K., Abraham, A., Gabralla, L. A., & Subramaniaswamy, V. (2022). Development of an end-to-end deep learning framework for sign language recognition, translation, and video generation. *IEEE Access*, 10, 104358-104374.
9. Patel, M. (2023). American sign language detection (Doctoral dissertation, California State University, Northridge).
10. Adeyanju, I. A., Bello, O. O., & Adegboye, M. A. (2021). Machine learning methods for sign language recognition: A critical review and analysis. *Intelligent Systems with Applications*, 12, 200056.
11. Wadhawan, A., & Kumar, P. (2020). Deep learning-based sign language recognition system for static signs. *Neural computing and applications*, 32(12), 7957-7968.
12. Sharma, S., Gupta, R., & Kumar, A. (2023). Continuous sign language recognition using isolated signs data and deep transfer learning. *Journal of Ambient Intelligence and Humanized Computing*, 1-12.
13. Kothadiya, D., Bhatt, C., Sapariya, K., Patel, K., Gil-González, A. B., & Corchado, J. M. (2022). Deesign: Sign language detection and recognition using deep learning. *Electronics*, 11(11), 1780.
14. Jia, W., & Li, C. (2024). SLR-YOLO: An improved YOLOv8 network for real-time sign language recognition. *Journal of Intelligent & Fuzzy Systems*, 46(1), 1663-1680.
15. Imran, A., Hulikal, M. S., & Gardi, H. A. (2024). Real Time American Sign Language Detection Using Yolo-v9. *arXiv preprint arXiv:2407.17950*.
16. Madanbhavi, L., Hireraddi, D., Nagaral, N., Patil, K., Dongrekar, G., Iraddi, M., ... & Patil, C. (2024, October). Sign Language Detection Using YOLOv10. In *2024 5th IEEE Global Conference for Advancement in Technology (GCAT)* (pp. 1-6). IEEE.
17. Alsharif, B., Alalwany, E., Ibrahim, A., Mahgoub, I., & Ilyas, M. (2025). Real-Time American Sign Language Interpretation Using Deep Learning and Keypoint Tracking. *Sensors*, 25(7), 2138.
18. Sundar, B., & Bagyammal, T. (2022). American sign language recognition for alphabets using MediaPipe and LSTM. *Procedia Computer Science*, 215, 642-651.
19. Subramanian, B., Olimov, B., Naik, S. M., Kim, S., Park, K. H., & Kim, J. (2022). An integrated mediapipe-optimized GRU model for Indian sign language recognition. *Scientific Reports*, 12(1), 11964.